Journal of Experimental Psychology: Applied
2004, Vol. 10, No. 1, 42–54

1076-898X/04/$12.00 DOI: 10.1037/1076-898X.10.1.42

Attending to the Execution of a Complex Sensorimotor Skill: Expertise Differences, Choking, and Slumps

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A simulated baseball batting task was used to compare the relative effects of attending to extraneous information (tone frequency) and attending to skill execution (direction of bat movement) on performance and swing kinematics and to evaluate how these effects differ as a function of expertise. The extraneous dual task degraded batting performance in novices but had no significant effect on experts. The skill-focused dual task increased batting errors and movement variability for experts but had no significant effect on novices. For expert batters, accuracy in the skill-focused dual task was inversely related to the current level of performance. Expert batters were significantly more accurate in the skill-focused dual task when placed under pressure. These findings indicate that the attentional focus varies substantially across and within performers with different levels of expertise.

What do athletes pay attention to when executing a complex sensorimotor act such as putting a golf ball or hitting a baseball? Theories of skill acquisition (Anderson, 1982, 1983; Fitts & Posner, 1967) suggest that the answer to this question depends largely on the athlete's level of expertise with the particular skill. For novice performers, it has been proposed that skill execution requires that attention be paid to each component stage of the motor act. At this level of performance (referred to as the *cognitive stage* or the *declarative stage*) it is assumed that skill execution depends on a set of unintegrated control structures that must be held in working memory and attended to in a step-by-step fashion. These attentional and memory requirements result in the slow, nonfluent, and error-prone movement execution that is the hallmark of a novice performer. As expertise develops through practice and the actor reaches the highest stage of skill execution (the *autonomous stage* or the *procedural stage*), it has been proposed that the role of attention in performance changes dramatically. As procedural knowledge develops, the conscious step-by-step control of execution is no longer required. Instead skill execution is assumed to operate by fast, efficient control procedures that can function largely without the assistance of working memory or attention. Therefore, the attentional mechanisms involved in the execution of a sensorimotor skill would seem to be one of the most prominent behaviors that distinguishes experts and novices.

These theories of skill acquisition make several predictions about how the attentional processes of novice and expert performers should differ, with many important implications for sports skills training. The main purpose of the present study was to explore the attentional mechanisms involved in baseball batting across different levels of skill expertise and to directly test some of the predictions of skill acquisition theory in this domain. The specific predictions and the results of previous experiments designed to evaluate them are described next.

Predictions From Cognitive Theories of Skill Acquisition

Prediction 1: Expert Performers Should Be Able to Process Extraneous Information More Effectively

In the procedural stage of skill acquisition it has been proposed that successful performance no longer requires that the performer attend step by step to each component of the motor act. Therefore, expert performers should have more attentional resources available to process sensory information that is extraneous to task completion.

Using a secondary task of visual shape identification, Leavitt (1979) was the first to demonstrate that the introduction of a secondary task hinders sensorimotor skill execution for novice performers but not expert performers. In this study, skating speed while stick handling through pylons was significantly slower under dual-task conditions for novice hockey players as compared with experts. Similar results have been found for soccer ball dribbling (Smith & Chamberlin, 1992) and ball catching (Parker, 1981). However, despite the large effects found in these studies, the manipulations used do not permit clear inferences about the attentional processes involved in skill execution. As discussed in detail by Beilock, Carr, MacMahon, and Starkes (2002), the use of a visual discrimination task is likely to have caused considerable structural interference (i.e., it may have interfered with detection of the sensory information required for the primary task [Wickens, 1980]). Because novice performers are more reliant on visual feedback (A. M. Williams, Davids, Burwitz, & Williams, 1992), differences in performance could be explained at the level of visual processing.

To reduce the problem of structural interference, a task that has been commonly used in dual-task experiments is the detection of a randomly presented auditory probe. Abernethy (1988) reported that novice badminton players made significantly more unforced

This work was supported by National Science Foundation Faculty Early Career Development Program Award 0239657. I thank Roger Schvaneveldt for advice on statistics.

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errors when required to detect an auditory probe as compared with single-task conditions. The probe detection task had no effect on expert players. The auditory probe reaction time technique has also been used to demonstrate expertise differences in the attentional processes involved in shooting a pistol (Rose & Christina, 1990) and receiving a volleyball serve (Castiello & Umilta, 1992). One problem associated with the use of the auditory probe reaction time technique is that it could be argued that simple detection of an auditory stimulus does not place substantial demands on attentional processing.

Beilock, Carr, and colleagues recently conducted a series of studies that more clearly elucidated how processing of extraneous secondary tasks varies as a function of expertise. Beilock, Wierenga, and Carr (2002) had novice and experienced golfers execute a primary task of putting while simultaneously performing a secondary task of monitoring a stream of auditory signals (either words or tones of different frequencies) for a particular target signal. They found that the mean putting error was not significantly different in single-task and dual-task conditions for expert golfers whereas the mean error was 4 cm greater in the dual-task than in the single-task conditions for novices. Auditory target detection failures were similar in the single-task and dual-task conditions for both groups. These findings suggest that the difference in putting performance between novice and expert golfers is due to the fact that experts have sufficient attentional resources available to perform the auditory monitoring without hindering putting performance whereas novices do not. In a separate study, Beilock, Carr, et al. (2002) found that this secondary auditory monitoring task had similar effects on the ability of expert and novice soccer players to dribble through a series of pylons. One limitation of these studies is that reaction times were not measured for the auditory monitoring task. Depending on the nature of the secondary task, it is common in dual-task experiments for observers to perform with a high degree of accuracy but with much slower reaction times (e.g., Gray, 2000). Thus, it is possible that the novice performers in Beilock et al.'s studies were actually performing worse on the secondary task but that this was not detected because participants traded off speed for accuracy.

Another way of measuring the attention devoted to an extraneous secondary task is to use a recognition memory test for nontarget items. Because attention at encoding is known to improve the explicit recall of information (Craik, Govoni, Naveh-Benjamin, & Anderson, 1996), experts should perform better on recognition tests than novices. Beilock et al. (2002) provided evidence for this proposal as expert golfers were better able to recognize nontarget words that were presented in an auditory stream than were novice golfers.

In real-world sports performance, an implication of the proposal that experts have more attentional resources available is that elite athletes should be better able to detect visual cues that are not directly related to the task they are performing but are important for overall success (e.g., detecting the position of one's teammates on the court while dribbling a basketball). Evidence for this has been presented in several studies (Gray, 2002a; A. M. Williams et al., 1992).

Prediction 2: Expert Performers Should Have Less Access to Information About Movement Execution

Current theories of skill acquisition propose that at expert levels of performance, athletes should have poorer ability to verbalize the stages involved in skill execution (i.e., "expertise-induced amnesia" [Beilock & Carr, 2001]) for two primary reasons. First, the procedural knowledge underlying expert-level performance is by definition less accessible to verbal recall than the declarative knowledge structures used by novices. Second, because less attention is devoted to skill execution, memory for its components should be worse for expert players.

Beilock, Carr, and colleagues (Beilock & Carr, 2001; Beilock et al., 2002) tested this prediction by comparing novice and expert golfers in the dual-task putting paradigm described above. After each block of 30 putts participants completed a questionnaire designed to assess their episodic recollection for the last putt they had executed (i.e., all of the steps they went through during the putt). The main finding was that the mean number of steps listed in the questionnaire was significantly lower for expert golfers than for novice golfers. One limitation of this recollection memory paradigm is that it only indirectly measures attention and knowledge during skill execution. It would seem that a more direct test of the idea that declarative knowledge is reduced in experts would be to compare expert and novice performers' abilities to make judgments about particular aspects of the motor control (e.g., speed of club movement, orientation of body).

Prediction 3: Attending to Movement Execution Should Degrade Performance for Experts

Using the golf-putting task described above, Beilock et al. (2002) introduced a secondary skill-focused task in which golfers were required to monitor the movement of the club head and say the word "stop" at the exact moment they finished their follow-through. Attending to the head of the putter in this manner significantly reduced the accuracy of putting relative to single-task performance for expert golfers. In a separate experiment, Beilock et al. (2002) used a similar manipulation to compare the effects of skill-focused attention on soccer dribbling. For this task participants were presented with tones at random intervals while they dribbled a ball through a slalom course. On hearing the tone, they were instructed to verbalize the side of their foot that was currently in contact with the ball (i.e., inside or outside). For experienced soccer players, the mean time to complete the dribbling course was significantly slower in this skill-focused condition than in an auditory word-monitoring dual-task condition. This effect occurred only when players dribbled with their dominant right foot; for left-foot dribbling, times were faster in the skill-focused condition. This latter finding suggests that attentional focus can have differential effects within an individual performer.

A limitation of both the golf and soccer experiments conducted by Beilock, Carr, et al. (2002) and Beilock, Wierenga, et al. (2002) is that performance on the secondary skill-focused task was not measured rigorously. In both experiments only the number of trials in which the participant failed to make a verbal response was recorded. Whether the golfers said "stop" at the actual end of their follow-through and whether soccer players could accurately report the side of the foot that was in contact with ball were not measured. Therefore, it is not clear how well participants could attend to movement execution and whether this ability varied with experience. Given the strong effect on the primary tasks in these experiments, it is clear that participants were devoting some attention to the skill-focused secondary tasks; however, it would be informative to a have a quantification of the attentional processing in these tasks.

Instead of using a secondary task that explicitly requires the performer to attend to skill execution, a more common approach has been to provide instructions that encourage a skill-focused attentional orientation (e.g., “Concentrate on the force applied by your feet” or “Be aware of the way that you throw the ball”). See Singer (1988) for a detailed example of this type of instruction. As discussed in a recent review of this body of research (Wulf & Prinz, 2001), instructions that encouraged skill-focused attention resulted in worse performance than instructions that encouraged an external focus (e.g., “Focus on the ball’s trajectory as it leaves the club”) and were often worse than no instructions at all (Wulf, Wiegelt, Poulter, & McNevin, 2002). This effect has been demonstrated for a wide range of perceptual–motor actions, including slalom skiing, tennis, and golf. A major limitation of this approach is that it is not clear to what extent performers actually used the attentional focus that was intended (i.e., formal manipulation checks are rarely used). It is also difficult to study the effects of expertise using this method because it seems unlikely that a highly skilled performer with a well-developed strategy would actually follow the experimental instructions.

It has been proposed that a more naturally occurring factor that may increase skill-focused attention is the psychological stress associated with performance in high-pressure situations (Baumeister, 1984; Carver & Scheier, 1981). Indeed, it has been demonstrated that the level of self-focused attention (assessed by questionnaire) increases as the time until a critical sporting event decreases (Liao & Masters, 2002). There has been considerable debate over the exact mechanisms responsible for performance decrement under pressure and the associated methods for its prevention. One theory is that the primary effect of pressure is to create a distracting environment that shifts the performer’s attention away from task-relevant information, similar to a dual-task experimental manipulation (Wine, 1971). The main alternative theory is that pressure serves to raise the amount of skill-focused attention, leading to a disruption of proceduralized knowledge (Baumeister, 1984). In a golf-putting study, Beilock and Carr (2001) directly tested these two hypotheses. Prior to performance under high pressure, golfers participated in either a dual-task training condition in which they were given practice at dealing with extraneous distractions (streams of words presented over a loudspeaker) or a self-consciousness training condition in which they were given practice at dealing with an environment that induced an increase in self-focused attention (being videotaped while putting). Consistent with the skill-focused attention theory, only the self-consciousness training eliminated choking under pressure (see also Lewis and Linder, 1997).

Masters (1992) proposed an extension of the self-focused attention theory of choking called the *reinvestment hypothesis.* It is argued that when under pressure performers reinvest the explicit or declarative knowledge acquired during the early stages of skill acquisition, leading to a disruption of procedural knowledge. The interesting prediction from this hypothesis is that training under a high degree of self-consciousness and skill focus (e.g., being videotaped) should lead to the development of a large amount of explicit, declarative knowledge that will increase the likelihood that performance will fail when under pressure. In support of this prediction, Liao and Masters (2002) recently demonstrated that self-focused training of basketball free-throw shooting led to a greater amount of explicit knowledge (e.g., rules) and worse performance under pressure relative to a control group. From these findings Liao and Masters made the somewhat counterintuitive proposal that the sensorimotor skills should be acquired with as little explicit knowledge as possible.

Aims of the Present Study

The main purpose of the present study was to examine the attentional mechanisms underlying the execution of baseball batting across different levels of expertise and different performance environments. This study extends previous work on attention and skill execution to a sensorimotor skill that is not self-paced. An additional goal of the present study was to measure the impact of attentional focus on movement kinematics to begin to understand how attentional mechanisms can disrupt physical performance processes. In Experiment 1, a modification of Beilock et al.’s (2002) dual-task paradigm was used to test Predictions 1–3 described above with experienced and novice baseball players. In Experiment 2, a dual-task paradigm was used to test whether the relationship between skill-focused attention and expert performance described in Prediction 3 also operated in the opposite direction (i.e., does the level of skill-focused attention depend on the current performance level?). In Experiment 3, a further test of Prediction 3 was conducted by introducing performance pressure on expert baseball players.

Experiment 1: Attention in Baseball Batting

Purpose and Rationale

The aims of Experiment 1 were threefold: (a) to compare the relative effects of attending to extraneous information and attending to skill execution on simulated baseball batting performance, (b) to compare how these effects differed as a function of expertise, and (c) to examine the effect of attentional focus on the kinematics of batting movements.

Method

Apparatus

The baseball batting simulation used in the present study has been previously described in detail (Gray, 2002a, 2002b). In brief, participants swung a baseball bat at a simulated approaching baseball. The simulated ball was an off-white sphere texture mapped with red laces. The background was black. The simulated ball was displayed on a 28-cm vertical × 36-cm horizontal SVGA monitor (Viewsonic Model PT795, Walnut, CA) that ran at 120 Hz. The monitor was viewed from a distance of 3.5 m in a dimly lit room. A sensation of motion toward the batter was created by increasing the angular size of the ball. The vertical position of the ball on the monitor was changed to simulate the drop of the ball as it approached the batter. The simulated pitcher could throw only two pitches (a fastball and a slow ball [i.e., change-up]). “Slow” pitches traveled at 70 ± 1.5 mph (31 ± 0.67 m/s), and “fast” pitches traveled at 85 ± 1.5 mph (38 ± 0.67 m/s). Speed was chosen randomly on each trial to be either fast or slow. The only force affecting the flight of the ball was gravity (e.g., the effects of air resistance and spin on the ball’s flight were ignored). The height of the simulated pitch $y(t)$ was changed according to

$$y(t) = -1/2 \times g \times t^2, \quad (1)$$

where g is acceleration of gravity (32 ft/s [9.8 m/s]). All pitches were strikes and traveled down the center of the plate; therefore, successful hitting required the batter to accurately estimate the point in time when the ball would cross the plate as well as its height above the ground at that instant in time. Although this is a simplified version of real baseball batting, in which there can also be a large variation in the lateral position of the ball when it crosses the plate, previous research has shown that the judgment of ball height is most difficult for the batter and is the cause of the majority of performance errors (Bahill & Karnavas, 1993; Gray, 2002a; Watts & Bahill, 1991). The relative difficulty in judging ball height becomes evident when the spatial dimensions of a bat and a ball are considered: For a typical baseball bat, the horizontal hitting area (i.e., the length of the bat) is roughly three times the diameter of the ball, whereas the vertical hitting area is roughly equivalent to the ball's diameter. Therefore, the simplified task used in the present study involved the most demanding perceptual judgment in batting.

Mounted on the end of the bat (Louisville Slugger Tee Ball bat; 25 in. long [63.5 cm]) was a sensor from a Fastrak (Polhemus, Colchester, VT) position tracker. The *x, y,* and *z* positions of the end of the bat were recorded at a rate of 60 Hz. These time series data were used to calculate the temporal accuracy of the swing as described below. A second sensor was mounted on top of the batter's lead foot (i.e., the one closer to the display when in the batting stance), and its position was also recorded at a rate of 60 Hz.

During all conditions participants wore head-mounted microphones (Shure Model SM10A, Niles, IL) to record vocal reaction times in response to auditory signals. Auditory signals were 80-ms duration tones presented via a loudspeaker. The tone frequency was either 250 or 500 Hz and was chosen randomly from trial to trial.

Procedure

Baseball batting task. The primary task for all conditions was that of hitting the virtual ball. Participants were instructed that the purpose of the study was to measure baseball batting performance. No instructions were given as to how hard to swing the bat. Ten seconds before the ball appeared on the screen, a progress bar that indicated the time until pitch release was presented on the monitor. The progress bar was used to give a crude simulation of the pitcher's windup. The progress bar disappeared 0.5 s before the pitch was released.

Visual feedback indicating the success of the batter's swing was given as follows: The *x*-, *y*-, and *z*-coordinates of the bat and ball as a function of time were used to estimate the point of contact on the bat (if there was contact) and bat speed. These variables were then used to approximate the trajectory and speed of the ball leaving the bat. Five seconds after the virtual ball crossed the plate, an image of a baseball diamond was presented on the monitor. A red line extending from home plate indicating the trajectory and distance of the ball was drawn on the diamond. If no contact was made, a large red *X* was presented to indicate a strike. Text messages were also displayed to indicate foul balls and home runs (balls estimated to travel further than 300 ft [91.5 m]).

The time interval between successive pitches was 20 s. Each block of trials consisted of 20 pitches. At the end of each block, a statistical summary showing the number of contacts, fair balls, and home runs was displayed on the monitor.

Attention conditions. A single-task condition (i.e., baseball batting only) was compared with two dual-task conditions: an extraneous dual-task condition in which participants were required to perform a simple auditory discrimination while batting and a skill-focused dual-task condition in which participants were required to make a judgment about the movement of the bat during swing execution. To allow for a direct evaluation of the effects of attention, the visual and auditory stimuli presented to the participant were identical in all experimental conditions. Specifics of the three task conditions are described below.

Extraneous dual-task condition. In this condition, participants executed the baseball batting task described above while simultaneously performing a two-alternative, forced-choice auditory frequency discrimination task. The sequence of events for a typical experimental trial is shown in Figure 1. At a random time between 300 and 800 ms after the virtual ball appeared on the screen, a single auditory tone was presented for a duration of 80 ms. The frequency of the tone was either 250 or 500 Hz and was randomly chosen from trial to trial. The participant's task was to signal the tone frequency by saying "high" or "low" into the head-mounted microphone as fast as possible. The reaction time and accuracy of this response were recorded. If the participant did not respond within 1.5 s after the onset of the tone, the trial was discarded. No feedback was given as to the accuracy of the frequency judgment.

Skill-focused dual-task condition. The batting task and auditory stimuli were identical to those described in the extraneous dual-task condition (as shown in Figure 1); the only difference was the judgment made by the participant. In this condition, the participant's task was to signal whether the bat was moving downward or upward at the instant the tone was presented by saying "up" or "down" into the microphone as fast as possible. Participants were instructed to respond in the same manner for both tone frequencies. Vocal reaction time and response accuracy (calculated using the vertical velocity of the bat) were again recorded. No feedback was given as to the accuracy of the skill-focused judgment. Trials in which the vertical velocity of the bat was zero at the onset of the tone were discarded.

Single-task condition. The two dual-task conditions were compared with a single-task condition in which participants were instructed to ignore the auditory tones (i.e., no secondary response was required) and to just complete the batting task.

All participants took part in all three experimental conditions. Each participant completed 100 trials (5 repeats of 20 trials) per condition. Before completing experimental trials, all participants completed three blocks of 20 trials of batting practice. All participants completed the single-task condition first followed by the two dual-task conditions. The

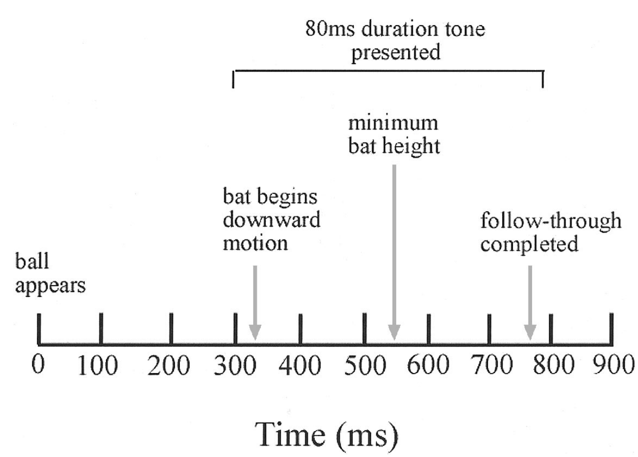

Figure 1. Sequence of events in a typical experimental trial. At a random time between 300 and 800 ms after the virtual ball appeared on the screen, a single auditory tone was presented for a duration of 80 ms. The frequency of the tone was either 250 or 500 Hz and was randomly chosen from trial to trial. In the single-task condition, hitters were instructed to ignore the tones. In the extraneous dual-task condition, participants were instructed to indicate the tone frequency by saying "high" or "low" into the head-mounted microphone as fast as possible. In the skill-focused dual-task condition, participants were instructed to indicate whether their bat was moving downward or upward at the instant the tone was presented by saying "up" or "down" into the microphone as fast as possible.

order of the dual-task conditions was counterbalanced across participants within each skill-level group.

Data Analysis

Figure 2 shows a typical recording of bat and ball height as a function of time. From these movement traces the following variables were calculated.

Mean temporal swing error (MTE). For this study, it was assumed that the batter was attempting to contact the ball when the bat was at its minimum height, as commonly instructed by batting coaches (T. Williams & Underwood, 1970). Therefore, the temporal error was defined as the difference between the time when the ball crossed the front of the plate and the time when the minimum bat height occurred (Gray, 2002a).

Onset of downward motion of the bat. This refers to the point in time when the bat began moving downward from the batter's shoulder. The criterion for this variable was four consecutive height samples that were lower than the previous sample.

Timing of lead-foot movements. One of the main motor-control timing mechanisms that batters use is to shift their weight toward the back leg by lifting the lead foot (i.e., the one closer to the oncoming ball) off the ground. The time at which the front foot broke contact with the ground was measured using the sensor mounted on the participant's foot.

Hitting kinematics were compared for the different attention conditions. Investigations of the coordinated movements involved in a baseball swing have revealed that hitting involves a complex chain of muscle activity (Shaffer, Jobe, Pink, & Perry, 1993; Welch, Banks, Cook, & Draovitch, 1995). A powerful swing requires precise timing between movements of the different body parts; therefore, for experienced hitters each event in the act of hitting (e.g., lead foot breaking contact with the ground, maximum shoulder rotation) occurs at roughly the same time before ball contact on each swing (Welch et al., 1995). In the present study, the ratio between the two movement components illustrated in Figure 3 was measured: (a) the windup—the time between when the batter's lead foot first breaks contact with the ground and when the foot reestablishes contact—and (b) the swing duration—the time between the first downward movement of the bat and the instant when minimum bat height occurs. This particular ratio was used for two primary reasons. First, previous anecdotal and experimental evidence has shown that the windup and swing duration are two of the primary aspects of a baseball swing that are adjusted from pitch to pitch by the batter (Gray, 2002a; Hubbard & Seng, 1954; T. Williams & Underwood, 1970). Second, these gross movements could be more readily detected with the motion tracking apparatus used in the present study as opposed to more subtle movements such as shoulder and hip rotations.

External validity. To evaluate the external validity of the virtual batting task used in the present experiments, the correlations between the main dependent variable (MTE) and two measures of real-word batting performance (batting average and on-base percentage) were measured. Statistics from the last full season played by each player were used. These analyses were performed only on expert players because these batting statistics were not available for the majority of novices used in the present study. The correlation between MTE and batting average was $-.71$ ($p < .05$), and the correlation between MTE and on-base percentage was $-.61$ ($p < .05$).

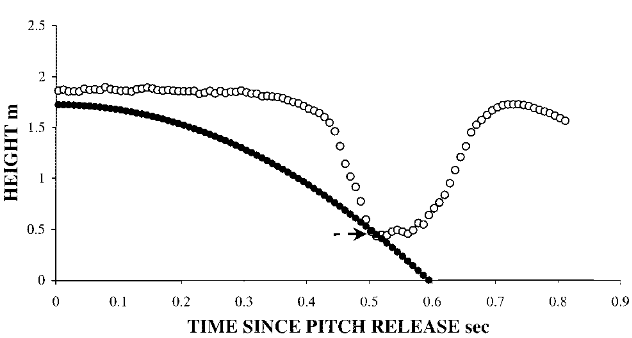

Figure 2. Example data record for one swing. Open circles plot the bat height as a function of time since release, and solid circles plot ball height. The arrow indicates the point at which the minimum bat height occurred. See text for details.

Participants

Twenty participants completed Experiment 1. *Expert batters* ($n = 10$) were Division 1–A college baseball players. The mean number of years of competitive playing experience for this group ranged from 12 to 19 ($M = 15.7$). *Novice batters* ($n = 10$) were participants who had played in recreational leagues within the last year and had less than 5 years of competitive playing experience. The mean number of years of competitive playing experience for this group ranged from 1 to 4 ($M = 2.3$). All participants were naive to the aims of the experiment and were paid a $10 hourly rate.

Results

Data Removal

From the 6,000 total trials, there were 84 in which the participants did not give a dual-task vocal response within the 1.5-s time period. In addition, there were 63 skill-focused dual-task trials in which the vertical velocity of the bat was zero at the onset of the tone. These 147 trials were discarded from the analyses. There was no significant difference between the number of trials discarded for novices and experts, $t(18) = 0.7$, $p > .05$, $d = 0.2$.

Batting Performance

Figure 4 plots the MTE for each of the three experimental conditions. It is clear from this figure that the attentional manipulations had differential effects on the two groups of batters: The extraneous dual task increased swing errors in novices but not in experts, whereas the skill-focused dual task increased swing errors in experts but not in novices. MTEs were analyzed using a 2×3 mixed factor analysis of variance (ANOVA) with expertise and attention condition as factors. As shown in Table 1, five planned orthogonal comparisons were made using the error term from the omnibus F test. Cohen's f statistic was used as an unbiased estimate of effect size. The main prediction for Experiment 1 was that the attentional manipulations would have differential effects on novice and expert baseball players. In support of this prediction, the comparison between the extraneous and skill-focused dual tasks across the different levels of expertise was significant. As would be expected if the simulated task used in the present study were related to real-world baseball performance, the MTE was significantly higher for novices than for experts. The effect sizes were large ($f > 0.4$) for both of these effects (Cohen, 1988). MTEs were also significantly higher in the extraneous dual-task condition than in the skill-focused dual-task condition.

Batting Kinematics

Table 2 shows the means and standard deviations of the windup/swing duration ratio for the two expertise groups in the three attention conditions. Again, there were substantial differences be-

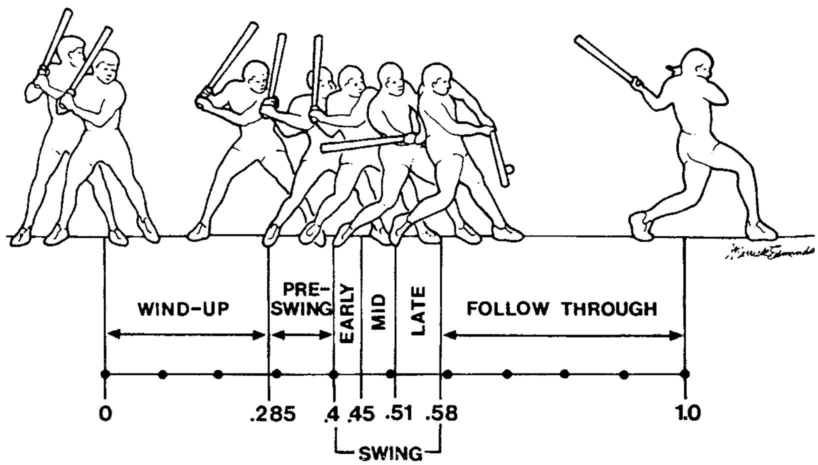

Figure 3. Phases of a baseball swing. Values are proportions of the total swing time. Modified from "Baseball batting: An electromyographic study," by B. Shaffer, F. W. Jobe, M. Pink, and J. Perry, 1993, *Clinical Orthopaedics and Related Research, 292,* Fig. 1. Copyright 1993 by Lippincott Williams & Wilkins. Adapted with permission.

tween experts and novices. These data were analyzed using the chi-square test for equal variances described by Snedecor and Cochran (1967). Two such tests were used to compare the mean variance of this ratio in the skill-focused dual task (s_f^2) with the mean variance of this ratio in the extraneous dual task (s_e^2). For expert batters, the null hypothesis that $s_f^2 \leq s_e^2$ was rejected, $\chi^2(9, N = 10) = 34.7$, $p < .05$, indicating that the variability of the windup/swing duration ratio was significantly higher in the skill-focused dual task than it was in the extraneous dual task. For novices, the null hypothesis that $s_e^2 \leq s_f^2$ was rejected, $\chi^2(9, N = 10) = 18.7$, $p < .05$, indicating that the variability of the windup/swing duration ratio was significantly lower in the skill-focused dual task than in the extraneous dual task.

Secondary Task Performance

The mean percentage of judgment errors for the dual-task conditions were as follows: expert/extraneous = 4.9% (*SD* = 2.4%),

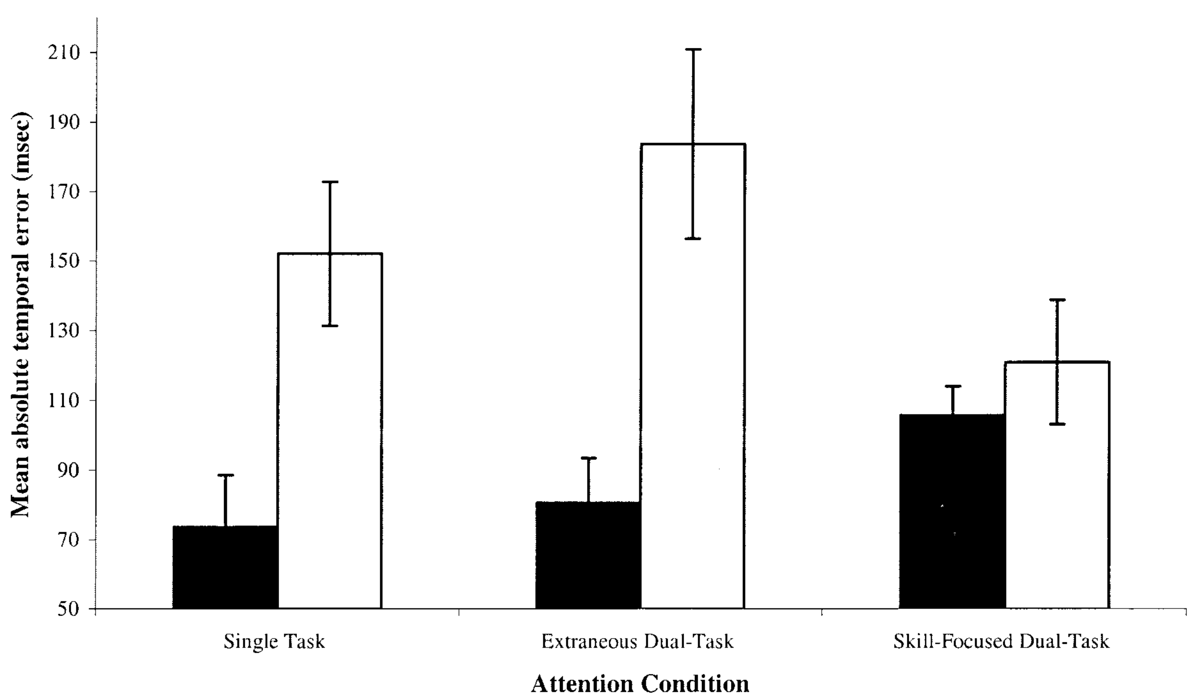

Figure 4. Overall batting performance in the three attention conditions used in Experiment 1. A: The mean absolute temporal error. Filled bars are for expert baseball players and open bars are for novice baseball players. Error bars are standard deviations.

Table 1
Mean Temporal Error Analysis of Variance for Experiment 1

Source and comparison	*df*	*MS*	*F*	Cohen's *f*
Expertise	1	64,419.27	89.11**	1.61
Participants/expertise	18	723.21		
Condition	2			
Single vs. dual	1	1,307.15	9.55**	0.20
Extraneous vs. skill	1	3,534.30	31.80**	0.45
Expertise × Condition	2			
Expertise × Single vs. Dual	1	1,242.71	9.08**	0.22
Expertise × Extraneous vs. Skill	1	19,272.52	173.39**	1.43
Single vs. Dual × Participants/Expertise	18	136.91		
Extraneous vs. Skill × Participants/Expertise	18	111.15		

** $p < .01$.

expert/skill-focused = 36.3% (*SD* = 9.6%), novice/extraneous = 7.7% (*SD* = 2.4%), and novice/skill-focused = 21.5% (*SD* = 6.3%). Performance for the skill-focused dual task was considerably worse than for the extraneous dual task. This effect was possibly due to a ceiling effect for the frequency discrimination judgment due to the large difference between the tone frequencies used. A 2 × 2 mixed factor ANOVA (results shown in Table 3) revealed a significant effect of expertise, with novices having a smaller percentage of secondary task errors than experts; a significant effect of attention condition; and a significant Expertise × Attention Condition interaction. This interaction occurred because experts made substantially more judgment errors in the skill-focused dual task than did novices.

The mean reaction times for the dual-task conditions were as follows: expert/extraneous = 395 ms (*SD* = 50), expert/skill-focused = 419 ms (*SD* = 33), novice/extraneous = 445 ms (*SD* = 39), and novice/skill-focused = 420 ms (*SD* = 54). A 2 × 2 mixed factor ANOVA (results shown in Table 4) revealed no significant effects of expertise or attention condition on reaction time; however, expertise and the Expertise × Attention Condition interaction did have medium effect sizes (Cohen, 1988).

Discussion

The results of Experiment 1 demonstrated significant differences in the attentional mechanisms of expert and novice baseball players. Experts in the present study appeared to be able to attend to a briefly presented auditory tone and to judge its frequency during swing execution without any effect on the temporal swing error. In contrast, introducing a secondary frequency discrimination task resulted in, on average, a 32-ms increase in the temporal swing error relative to single-task conditions for novice batters. This increase in temporal error was nearly four times the ±9-ms estimated temporal margin for error in baseball batting (Watts & Bahill, 1991). Clearly, unlike more experienced players, novices do not seem to have sufficient available attentional resources to simultaneously hit and attend to extraneous sensory information. This pattern of results provided support for Prediction 1 (i.e., that expert performers should be able to process extraneous information more effectively). In previous studies using the same simulated batting task, it has been reported that swings made by expert batters were more influenced by indirect sources of information (e.g., the direction of rotation of the ball, the pitch count, and the pitch sequence) than swings made by less experienced batters (Gray, 2002a, 2002b). The results of the present study suggest that a reason for this effect may be that less experienced batters do not have the available attentional resources needed to pick up these sources of information.

The influence of expertise was different in the skill-focused condition in Experiment 1. When given a secondary task that required continuous attention to a component of swing execution (i.e., the direction in which the bat was moving), batting performance was degraded for expert baseball players: on average, a 31-ms increase in temporal error relative to single-task performance. The results of the kinematic swing analyses suggest that this performance degradation is at least partially due to the fact that skill-focused attention in experts interfered with the sequencing and timing of the different motor responses involved in swinging a baseball bat. These findings provide support for Prediction 3 of skill acquisition theory (i.e., that attending to movement execution should degrade performance for experts).

Secondary task performance also differed for expert and novice batters. Consistent with Prediction 2 (i.e., that expert performers should have less access to information about movement execution), expert baseball players in the present study made signifi-

Table 2
Windup/Swing Duration Ratios (With Standard Deviations) for Experiment 1

Group	Single task	Extraneous dual task	Skill-focused dual task
Expert	0.13 (0.05)	0.14 (0.03)	0.28 (0.04)
Novice	0.46 (0.08)	0.55 (0.07)	0.38 (0.08)

Table 3
Percentage Accuracy Analysis of Variance for Experiment 1

Source	*df*	*MS*	*F*	Cohen's *f*
Expertise	1	369.31	11.94**	0.51
Participants/expertise	18	30.92		
Condition	1	5,124.61	124.78**	1.12
Expertise × Condition	1	782.34	19.05**	0.73
Condition × Participants/Expertise	18	41.07		

** $p < .01$.

Table 4
Mean Reaction Time Analysis of Variance for Experiment 1

Source	*df*	*MS*	*F*	Cohen's *f*
Expertise	1	6,375.21	4.39	0.27
Participants/expertise	18	1,452.27		
Condition	1	4.23	0.01	0.03
Expertise × Condition	1	5,736.01	2.26	0.32
Condition × Participants/Expertise	18	2,538.54		

cantly more errors when judging the direction of movement of their bat than did novices.

Experiment 2: Current Performance Level and Attentional Focus

Purpose and Rationale

In Experiment 1, skill-focused attention during baseball batting was artificially increased using a secondary task that required a judgment about swing execution. As discussed above, in explicit monitoring theories of "choking under pressure" it has been proposed that performance pressure can naturally increase the amount of skill-focused attention (Baumeister, 1984). In these theories it is assumed that the pressure increases self-consciousness and performance anxiety, causing the athlete to take conscious control over a proceduralized skill. What other situations might cause this to happen? Coaches commonly speak of athletes "pressing" or "thinking too much" in the midst of a period of subnormal performance (a slump) and, alternatively, being "unconscious" or "in the zone" when in the midst of a period of supernormal performance (a hot streak; Lau, Glossbrenner, Larussa, & Salzberg, 1992; T. Williams & Underwood, 1970). These observations suggest that there may be an inverse relationship between the current level of performance and the amount of skill-focused attention. In Experiment 2, this prediction was tested explicitly.

Method

Apparatus and Procedure

The method used in Experiment 2 was identical to that described in Experiment 1 except for the following: Participants were required to give dual-task responses on randomly selected trials rather than giving them for every trial in an experimental block. Furthermore, dual-task responses occurred after the swing was completed rather than during swing execution. This randomized after-swing judgment method was used as opposed to the procedure used in Experiment 1 to ensure that participants could not use a different attentional strategy for the different secondary tasks. On each trial the sequence of stimulus events occurred as shown in Figure 1. In some trials, one of two text prompts (*Frequency* or *Swing*) was presented on the screen 10 s after the onset of the tone (i.e., well after the swing was completed). Participants were instructed that on trials for which the *Frequency* prompt appeared they were to indicate whether the tone presented during the trial was the high-frequency tone or the low-frequency tone by saying "high" or "low" into the head-mounted microphone. In trials in which the *Swing* prompt appeared they were instructed to indicate whether their bat was moving downward or upward at the instant the tone was presented by saying "up" or "down" into the microphone. In trials for which no prompt appeared they were instructed that no response was required. Participants were further instructed that they had as long as necessary to make a response. Response reaction times were not measured in Experiment 2. Prompts were always presented after the hitting feedback. The text prompt displayed in each trial was chosen randomly from the three alternatives (i.e., *Swing*, *Frequency*, or no prompt). All trials in which the vertical velocity of the bat was zero at the onset of the tone were discarded. All participants completed 25 blocks of 20 trials.

Data Analysis

The main purpose of Experiment 2 was to compare the relationship between the current level of performance and the degree of skill-focused attention. The current level of batting performance was quantified in the following manner. For each hitter, Trials 25 through 500 were analyzed. For each of these trials the total number of hits (i.e., balls hit into fair play) in the previous 24 trials was calculated and used as a measure of current performance level. Blocks of 24 trials were chosen to roughly match the amount of trials used in previous studies of streaks in skilled performance (Gilovich, Vallone, & Tversky, 1985). The degree of skill-focused attention was quantified by using the number of judgment errors in the skill-focused secondary task (i.e., the *Swing* prompt). All trials for which the total number of hits was the same on the previous 24 trials were grouped, and the mean percentage judgment error for the secondary task was calculated for that group. For comparison, this analysis was also performed for the extraneous dual-task judgment. The number of hits was used instead of the MTE measure used in Experiment 1 because a discrete performance variable was needed to group the data for this analysis. In previous research using this simulated batting task a high correlation between the number of hits and the MTE ($r = .86$, $p < .05$) was reported (Gray, 2002a), suggesting that these two variables provide comparable measures of performance.

On the basis of the observations of hitting coaches (Lau et al., 1992; T. Williams & Underwood, 1970), it was predicted that there would be a significant positive correlation between the mean number of hits in the previous 24 trials and the mean number of errors in the skill-focused judgment, indicating a decrease in skill-focused attention at high levels of performance. Because it has been shown that attention to extraneous cues is associated with a higher level of batting performance (Gray, 2002a), it was predicted that the mean number of errors for the extraneous judgment would be negatively correlated with the mean number of hits in the previous 24 trials.

Participants

In Experiment 2 only expert baseball players were used. The participants were the 10 expert hitters that completed Experiment 1. Experiments 1 and 2 were conducted on separate days, and participants were allowed to rest between runs to avoid any fatigue effects.

Results

Data Removal

From the 5,000 total trials, there were 52 in which the vertical velocity of the bat was zero at the onset of the tone. These trials were discarded from the analysis.

Batting Performance

The mean number of hits for the three different prompts used in Experiment 2 were as follows: single task (i.e., no prompt) = 34.2 ($SE = 4.2$), extraneous dual task = 36.7 ($SE = 2.1$), and skill-focused dual task = 32.2 ($SE = 5.1$). A one-way ANOVA performed on these data revealed no significant effect for prompt type on the mean number of hits, $F(2, 18) = 1.2$, $p > .05$, $f = 0.13$. This result was not surprising given that the task prompt was presented after the swing was completed.

Secondary Task Performance

There was a significant positive correlation ($r = .89$, $p < .01$) for the skill-focused dual task. The correlation for the extraneous dual task ($r = -.37$, $p > .05$) was not significant.

Discussion

The results of Experiment 2 provided the first demonstration that normal variations in performance level may be associated with a continuum between high and low levels of skill-focused attention. When the mean performance for the batters in Experiment 2 was poor (i.e., a batting slump), the number of errors in the skill-focused judgment was lower than when the mean performance level was high (i.e., a hot streak). This effect is quite different from previous theories that have proposed that skill-focused attention primarily occurs in experts under a high degree of pressure (Baumeister, 1984; Masters, 1992).

The predicted negative correlation between extraneous task performance and number of hits was not found in Experiment 2. As discussed above, this lack of relationship could have been due to a ceiling effect on the tone frequency judgment.

Experiment 3: Choking Under Pressure and Attentional Focus

Purpose and Rationale

Two primary theories have been put forth to explain the phenomenon of choking under pressure. Wine (1971) proposed that pressure essentially serves as a distractor that draws the performer's attention away from the perceptual information needed for skill execution. Alternatively, Baumeister (1984) proposed that the main effect of pressure is to increase the amount of skill-focused attention, leading to a disruption of proceduralized processes. In previous research these two hypotheses have primarily been evaluated using training paradigms designed to give the performers experience in handling high levels of skill-focused attention and/or high levels of distraction prior to facing the pressure situation. Although it has been demonstrated that skill-focused training can reduce choking under pressure (Beilock & Carr, 2001; Lewis & Linder, 1997), which supports Baumeister's theory, it has also been shown that this type of training can increase choking (Liao & Masters, 2002). The purpose of Experiment 3 was to further study the mechanisms underlying choking under pressure using the dual-task baseball batting paradigm described in Experiment 2. If, as proposed by Baumeister, pressure increases the amount of skill-focused attention, then accuracy in the skill-focused dual task used in the present study should improve under conditions of performance pressure relative to low-pressure conditions.

Method

Apparatus and Procedure

The method used in Experiment 3 was identical to that described in Experiment 2 except for the following: Participants completed two blocks of 200 trials, a low-pressure pretest followed by a high-pressure posttest. The low-pressure pretest was identical to that described in Experiment 2. The pressure manipulations used were similar to those used by Beilock et al. (2002) for golf putting. For the high-pressure posttest, participants were instructed that they would be teamed with 1 other participant in the study and that if they and their teammate could increase their total number of hits by 15% they would receive a $20 reward. Furthermore, the participants were instructed that their teammate had already successfully raised his total number of hits by 15%. After completion of the experiment all participants were given the reward regardless of whether they met this performance criterion. Data from this group of participants were compared with a control group that performed the pretests and posttests without the pressure manipulation.

Participants

Twelve college Division 1A baseball players participated in Experiment 3. None of these participants completed Experiment 1 or 2. The mean number of years of competitive playing experience for this group ranged from 14 to 18 ($M = 16.1$). Participants were randomly assigned to either the pressure group or the control group.

Results

Data Removal

From the 4,800 total trials, there were 39 in which the vertical velocity of the bat was zero at the onset of the tone. These trials were discarded from the analysis. There was no significant difference between the number of trials discarded for the pressure group and the control group, $t(10) = 0.2$, $p > .05$, $d = 0.1$.

Batting Performance

The MTEs for Experiment 3 were as follows: pressure group/pretest = 82.0 ($SD = 9.1$), pressure group/posttest = 109.5 ($SD = 5.1$), control group/pretest = 89.6 ($SD = 6.6$), and control group/posttest = 85.1 ($SD = 7.6$). These data were analyzed using a 2 × 2 mixed factor ANOVA with group (pressure vs. control) and test interval (pre vs. post) as conditions. Table 5 shows the results of this analysis. There was a significant Group × Test Interval interaction, indicating that the pressure manipulation used in this study degraded batting performance (i.e., performers in the pressure group exhibited choking). This interaction had a large effect size. None of the participants in the pressure group succeeded in increasing batting performance by the criterion level of 15%.

Secondary Task Performance

Figures 5A and 5B plot the mean percentage of errors for dual-task judgments in the pretest and the high-pressure posttest for the pressure group and the control group, respectively. These data were analyzed using a 2 × 2 × 2 mixed factor ANOVA with group, dual-task condition, and test interval as factors. Table 6 shows the results of this analysis. Consistent with the results of Experiments 1 and 2, there was a significantly higher percentage of judgment errors for the skill-focused dual task than for the extraneous dual task. There was a significantly lower percentage of

Table 5
Mean Temporal Error Analysis of Variance for Experiment 3

Source	*df*	*MS*	*F*	Cohen's *f*
Group	1	470.83	9.06*	0.82
Participants/group	18	51.96		
Test interval	1	721.60	13.65**	0.97
Group × Interval	1	1,648.32	31.17**	1.23
Interval × Participants/Group	18	52.88		

* $p < .05$. ** $p < .01$.

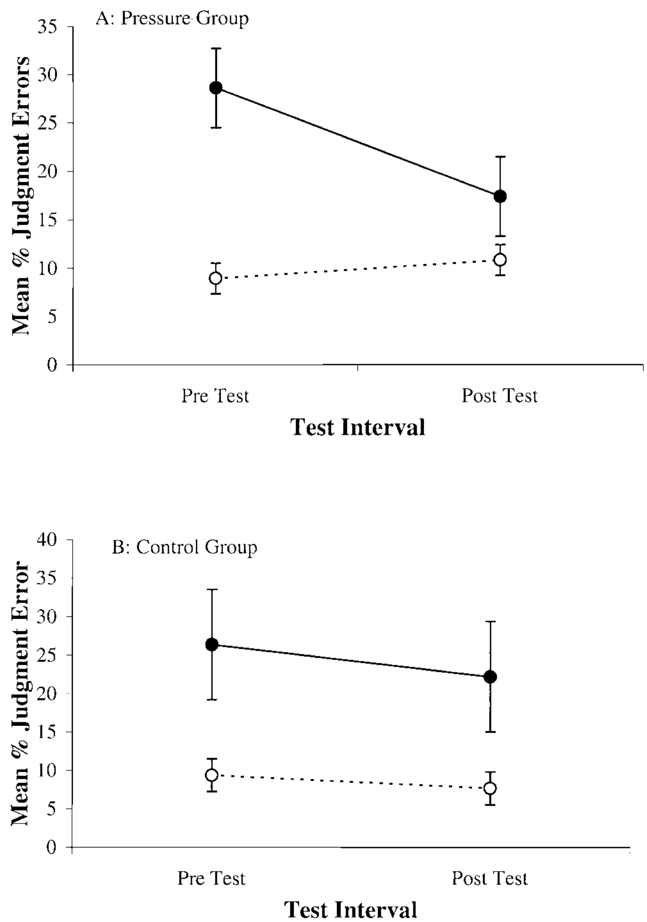

Figure 5. Mean percentage judgment errors for dual-task judgments in the pretest and posttest conditions of Experiment 3. A: Pressure group. B: Control group. Solid symbols and solid lines plot data for the skill-focused dual task (i.e., the "Swing" prompt), and open symbols and dashed lines plot data for the extraneous dual task (i.e., the "Frequency" prompt). Error bars are standard deviations.

judgment errors in the posttest than in the pretest. This effect can be best understood by examining the significant three-way Group × Dual-Task Condition × Test Interval interaction. As seen in Figure 5, the primary result of Experiment 3 was that the addition of performance pressure reduced the percentage of judgment errors in the skill-focused task for the pressure group but had no substantial effect on judgment errors in the extraneous dual task.

Batting Kinematics

To examine the effect of pressure on batting kinematics, the standard deviation of the windup/swing duration ratio was calculated as described in Experiment 1. Mean values of the standard deviation of this ratio were as follows: pressure group/pretest = 0.14 (SD = 0.04), pressure group/posttest = 0.39 (SD = 0.04), control group/pretest = 0.14 (SD = 0.04), and control group/posttest = 0.15 (SD = 0.03). The primary effect was that the introduction of performance pressure in the posttest led to increased variability in the windup/swing duration ratio. These data were again analyzed using the chi-square test for equal variances described above. Two such tests were used to compare the mean variance of this ratio in the pretest (s^2_{pr}) with the mean variance of this ratio in the posttest (s^2_{po}). For the pressure group, the null hypothesis that $s^2_{po} \leq s^2_{pr}$ was rejected, $\chi^2(9, N = 10) = 69.8$, $p < .05$. For the control group, the null hypothesis that $s^2_{po} \leq s^2_{pr}$ could not be rejected, $\chi^2(9, N = 10) = 10.3$, $p > .05$.

Discussion

The decrease in response errors for the skill-focused dual-task condition in Experiment 3 provided the first direct evidence that the level of skill-focused attention is significantly greater in expert performers when placed under pressure. As observed in Experiments 1 and 2, this change in attentional focus was associated with a degradation in batting performance: on average, a 28-ms increase in the temporal swing error relative to the low-stress pretest. The observed increase in skill-focused attention induced by performance pressure was also associated with changes in batting kinematics as the timing of the different stages of swing became more variable. These effects are discussed in more detail below.

One point that should be emphasized about the results of Experiment 3 is that performance pressure did not result in a complete deterioration of performance. In fact, participants in the present study actually performed significantly better on the skill-focused judgment task when under pressure. This finding suggests that choking under pressure is not due to an overloading of attentional capacity caused by the distracting effects of pressure as proposed by Wine (1971) but is caused by an inward shift of attentional focus as proposed by Baumeister (1984).

General Discussion

Differences in Attentional Processing as a Function of Expertise

Researchers, coaches, and athletes have long strived to find the key psychological characteristics that separate novice and expert performers. Theories of skill acquisition have proposed that the role of selective attention in skill execution may be one of the most crucial factors. The findings of Experiment 1 demonstrated signif-

Table 6
Percentage Judgment Error Analysis of Variance for Experiment 3

Source	*df*	*MS*	*F*	Cohen's *f*
Group	1	0.06	0.01	0.02
Participants/group	10	35.62		
Dual-task condition	1	2,501.73	498.35**	2.5
Group × Dual	1	21.14	4.21	0.23
Dual × Participants/Group	10	5.02		
Test interval	1	174.30	21.18**	0.11
Group × Test	1	8.51	1.03	
Test × Participants/Group	10	8.23		
Dual × Test	1	182.25	35.59**	0.28
Group × Dual × Test	1	85.28	16.66**	0.62
Dual × Test × Participants/Group	10	5.12		

** $p < .01$.

icant differences in attentional processing between baseball players of different levels of expertise. On the one hand, expert batters are better able to devote their attention to extraneous stimuli in the environment during skill execution than are novices. It is perhaps these available attentional resources that allow top-level players to use information that is not directly required for skill execution (e.g., the position of the infielders, the pitcher's body language) to further enhance batting success (Gray, 2002b). On the other hand, experts are less efficient at attending to and making judgments about the online execution of a skill than are novices. Experts in the present study were less accurate in making the skill-focused judgment, but also, in attempting to do so, batting performance was substantially degraded. These results are consistent with the idea that highly skilled performance is based on proceduralized knowledge structures (Anderson, 1982).

The results of Experiment 1 complement the experiments by Beilock, Carr, and colleagues (Beilock & Carr, 2001; Beilock et al., 2002) on golf putting and soccer dribbling that found similar expert–novice differences in attentional processing. One significant difference between the present study and this previous research is that baseball batting is not a self-paced task. Because of the severe temporal constraints involved (a 95-mph [42.5 m/s] major league pitch reaches the plate in less than 500 ms), a batter has little choice regarding when to sample the environment for perceptual information and when the different stages of the swing will be initiated. Conversely, in self-paced tasks it has been shown that the kinematics of skill execution can be adjusted on the basis of the frequency at which perceptual information is made available (Amazeen, Amazeen, Post, & Beek, 1999). Therefore, one might expect very different attentional-processing characteristics for self-paced and non-self-paced tasks given that the opportunity for attentional task switching should be considerably less for a task that is not self-paced. However, results of the present study demonstrated similar effects of dual-task manipulations on batting performance as found for golf putting. Another important difference between the present study and previous research is that the accuracy of the skill-focused judgment was measured. This measurement provided the first evidence that experts have less direct real-time access to knowledge about skill execution than do novices.

Variations in Attentional Processing in Expert Performers

The results of Experiment 2 demonstrated that there are substantial situational variations in attentional processing in expert-level performers. In other words, a high level of skill is not characterized by only one type of attentional allocation. When engaged in an extended period of below average performance, expert baseball players appear to increase the amount of skill-focused attention in an attempt to gain access to step-by-step execution of the swing. It is through this controlled cognitive modification of the component steps of skill execution that the batter attempts to break out of the performance slump (Taylor, 1988). Conversely, when the batter achieves a high level of skill, skill-focused attention appears to decrease substantially, and performance becomes largely proceduralized. Furthermore, as evident from the skill-focused judgment data reported in Experiment 2, skill-focused attention is not an all-or-nothing phenomenon, as the knowledge about movement execution was negatively and roughly monotonically related to the current level of performance.

Whether the application of skill-focused attention to movement execution is an effective strategy for an expert to break out of a performance slump is not clear. However, as seen in the results of Experiment 1, although this strategy will likely hurt current real-time performance in the short term, it may serve to improve skill execution in the long run. As suggested by Beilock et al. (2002), attention to skill execution in experts may be crucial for breaking down, altering, and adapting proceduralized knowledge that the performer has judged to be unproductive on the basis of cognitive self-regulation of his actions. In particular, this strategy would be most effective in practice situations in which the goal is not to maximize performance outcomes. Perhaps skill-focused attention in experts should no longer be considered a negative trait that must be avoided at all costs (Masters, 1992).

A dominant idea in skill acquisition theory has been that there are distinct, independent stages that a performer passes through as practice accumulates (Anderson, 1982; Fitts & Posner, 1967). Once the highest stage of skill acquisition is achieved, it has been proposed that only under the most severe conditions (e.g., extreme stress and pressure) will the performer regress to an earlier stage (A. M. Williams, Davids, & Williams, 1999). The results of the present study suggest that the movement from the cognitive stage, through the associative stage, to the procedural stage may not be so unidirectional. Instead expert performers may continuously cycle back and forth between these stages depending on the current level at which they are performing. This can be seen in the strong linear relationship between the accuracy of skill-focused judgments and performance. Perhaps it is as important for an athlete to learn strategies for moving quickly and effectively from the cognitive to procedural stage (i.e., techniques for acquiring new procedural knowledge) as it is to achieve that level in the first place.

It has long been suggested that choking in elite performers may result from a change in attentional allocation (Baumeister, 1984). However, because of the lack of manipulation checks in previous research, a direct link between skill-focused attention and performance degradation under pressure has yet to be established. In Experiment 3 of the present study a clear relationship between the accuracy of judgments about movement execution and performance error induced by a high-pressure environment was found, indicating an increase in skill-focused attention. It should be emphasized that in Experiment 3 the skill-focused judgment (a) was secondary to the task of hitting (no feedback was given, and participants were not required to meet a criterion level of accuracy), (b) was intermixed with other conditions so that the batter did not know if a skill-focused judgment was required at the point in time when the swing occurred, and (c) was completed after the swing was finished. Therefore, it would seem that the observed increase in skill-focused attention was not solely due to the use of a dual-task paradigm.

Previous research on skilled performance has not specifically addressed the issue of why a self-focused attentional orientation degrades performance in experts. This is an important omission because if the specific problems involved are not understood, a coach can do little more than instruct an athlete facing a high-pressure situation to "stop thinking about it and just do it." The results of the present study point to some mechanisms that may mediate this relationship.

The present study found significant changes in movement kinematics associated with an increase in skill-focused attention, in particular an increased variability in the relative timing of the different stages of a baseball swing. An increase in movement variability is what would be expected when a performer shifts control from encapsulated procedures to step-by-step cognitive control. Because the latter type of control involves working memory and selective attention it is subject to a greater amount of noise due to response reaction times, memory retrieval delays, and distraction. In baseball batting this increase in variability is problematic because it disrupts the kinetic link between the movement of the arms, shoulders, trunk, and legs that is required for a smooth and powerful swing (Welch et al., 1995). Although this effect seems to be in line with the predictions of skill-acquisition theory, on the surface it conflicts with previous research that has demonstrated a decrease in movement variability under conditions of performance pressure (Collins, Jones, Fairweather, Doolan, & Priestley, 2001; van Loon, Masters, Ring, & McIntyre, 2001). In these previous studies cross-correlations between the movements of pairs of joints (e.g., a shoulder and a hip) revealed a reduced variance (i.e., the movements of the joints are more tightly coupled) under pressure. This pattern of behavior is consistent with a novice strategy referred to as "freezing degrees of freedom" (Vereijken, Vanemmerik, Whiting, & Newell, 1992), which simplifies the problem of movement control by effectively reducing the number of movement parameters. The discrepancy between the effects observed in the present study and those observed in previous studies could be related to dependent variables used, as both the present kinematic results and the findings of Vereijiken et al. are consistent with a shift to a lower, less experienced level of performance.

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Received May 8, 2003
Revision received November 3, 2003
Accepted November 4, 2003 ■